Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. COMPUTER SCIENCE AND ENGINEERING

END SEMESTER MAKE-UP EXAMINATIONS, DECEMBER 2019

SUBJECT: DISTRIBUTED AND CLOUD COMPUTING [CSE 4102]

REVISED CREDIT SYSTEM
26.12.2019

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL FIVE** questions.
- Missing data may be suitable assumed.

1A. Define marshalling and construct CORBA CDR message using the following CORBA IDL for Person structure with an example. **4M**

```
struct Person{
        String name;
        String place;
        Unsigned long year;
};
```

1B. Explain in detail the design requirement of distributed systems. **3M**

1C. Explain in detail the different failure models of a distributed system with a necessary diagram. **3M**

2A. What are the different choices of RMI invocation semantics? Explain in detail how different delivery guarantees are assured using these invocation semantics? **4M**

2B. Explain with a neat diagram, how concurrency and collaboration is achieved in distributed mutual exclusion algorithm? **3M**

2C. What is the problem encountered in Lamport's logical clocks? Explain in detail, how this is solved using Vector clocks algorithm? **3M**

3A. There are 8 processes in a group numbered from 0 to 7. Process 4 noticed the crash of process 7. Explain the method of electing new coordinator using bully algorithm in the given scenario with a neat diagram. **4M**

3B. Explain the following client centric consistency model with a suitable example and a necessary diagram. **3M**

a) Monotonic Write
b) Read Your Write
c) Write Follows Read

3C.	What is cloud computing? What are the different features of cloud computing model which helps to enable the services and expectations of consumers?	**3M**
4A.	Explain Client Caching and Server Caching in Sun Network File System.	**4M**
4B.	Describe the challenges and risks associated with Cloud Computing services.	**3M**
4C.	With the help of neat labeled diagrams explain the life cycle of a Virtual Machine and the steps involved in provisioning a virtual server in a cloud computing environment.	**3M**
5A.	Explain the following with respect to Hadoop Distributed File System a) Heartbeat Signal and Block Report b) Checkpoint and Checkpoint Node c) Snapshot and BackupNode d) Balancer and Decommisioning	**4M**
5B.	Explain Map reduce programming model with a word count example.	**3M**
5C.	What are the different phases of SLA management in cloud?	**3M**

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